PV Diagram of an SI Engine

AUTOMOBILE ENGINEERING

B.TECH(MECHANICAL ENGINEERING)

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OBJECTIVE

To measure and plot the pressure vs volume diagram as a function of crank position for a four-stroke single-cylinder SI engine.

APPARATUS/SETUP REQUIRED:

4-Stroke SI Engine Test Rig

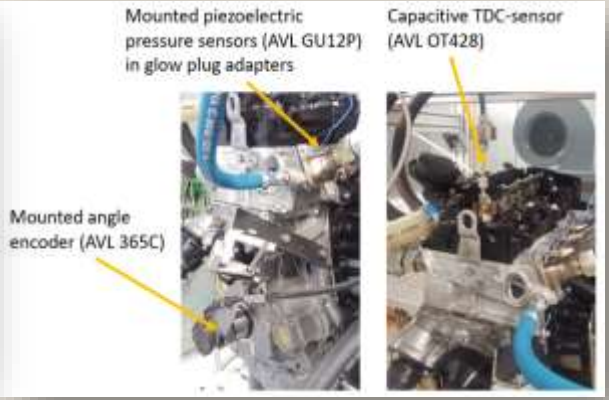
Single cylinder / Multi-cylinder

Pressure transducer

In-cylinder Pressure Sensor, mounted on cylinder head

Crank angle encoder / TDC sensor

Gives crank angle position (0–720°) and reference of TDC



Exhaust system

With Silencer

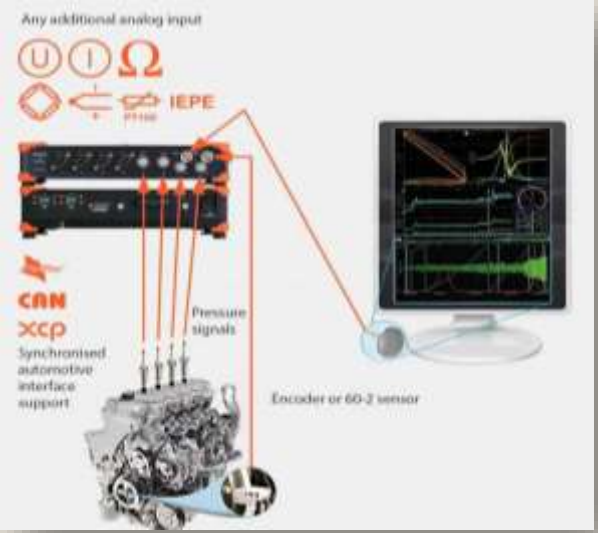
Tachometer

Used for engine speed measurement

Data acquisition system (DAQ)

Single cylinder / Multi-cylinder

Computer with analysis software (MATLAB / LabVIEW / Engine analysis software)

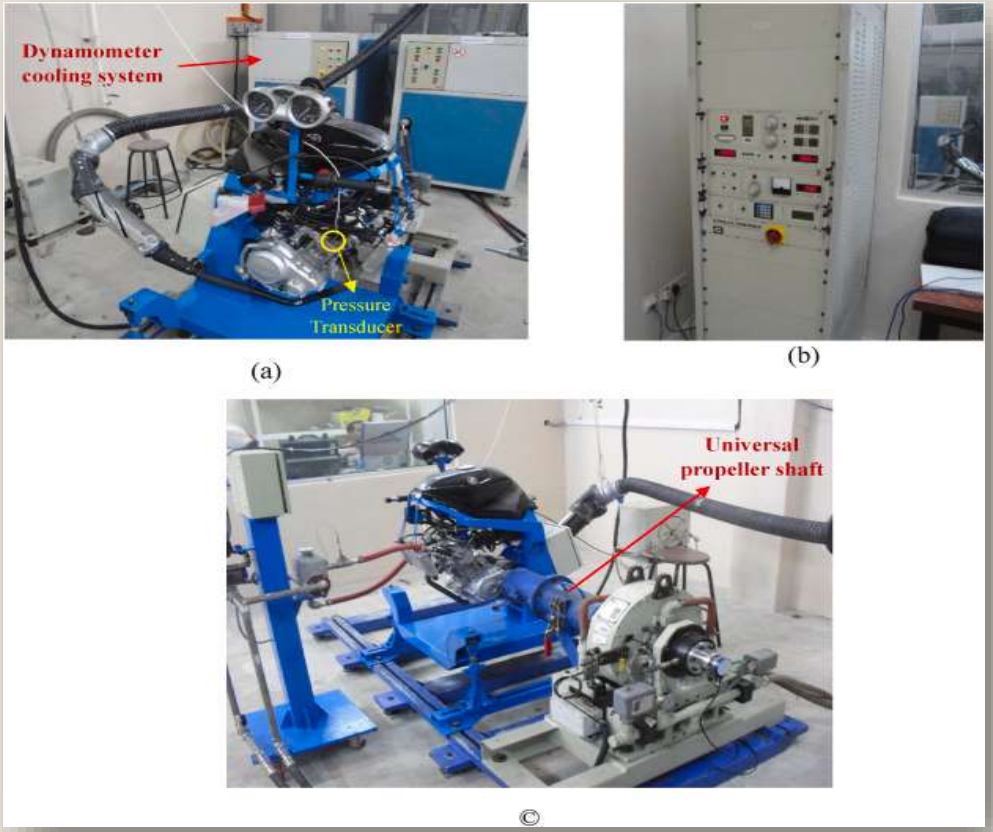


Charge amplifier / Signal conditioner: Converts sensor output to usable voltage

Fuel supply system: Petrol tank, fuel lines, carburetor/injector

Ignition system: Spark plug, ignition coil, battery

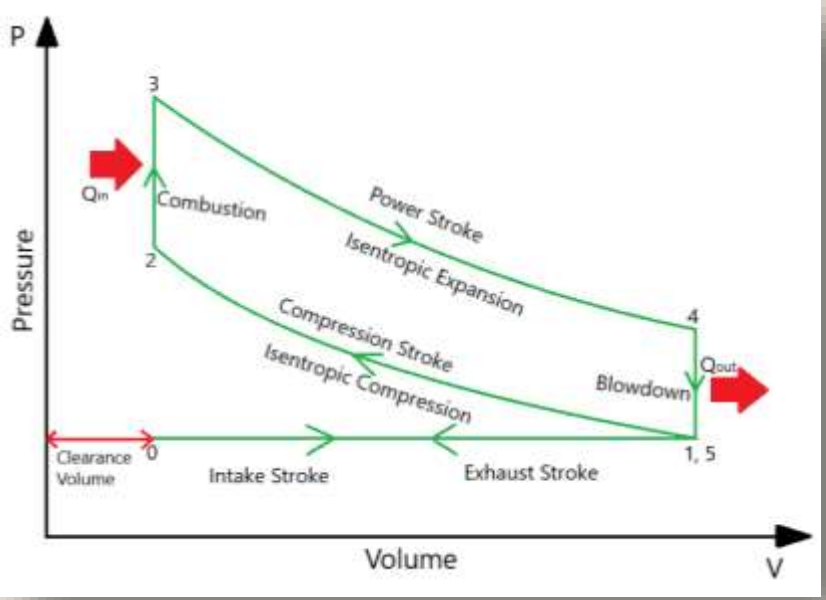
Cooling System: Air-cooled or water-cooled arrangement

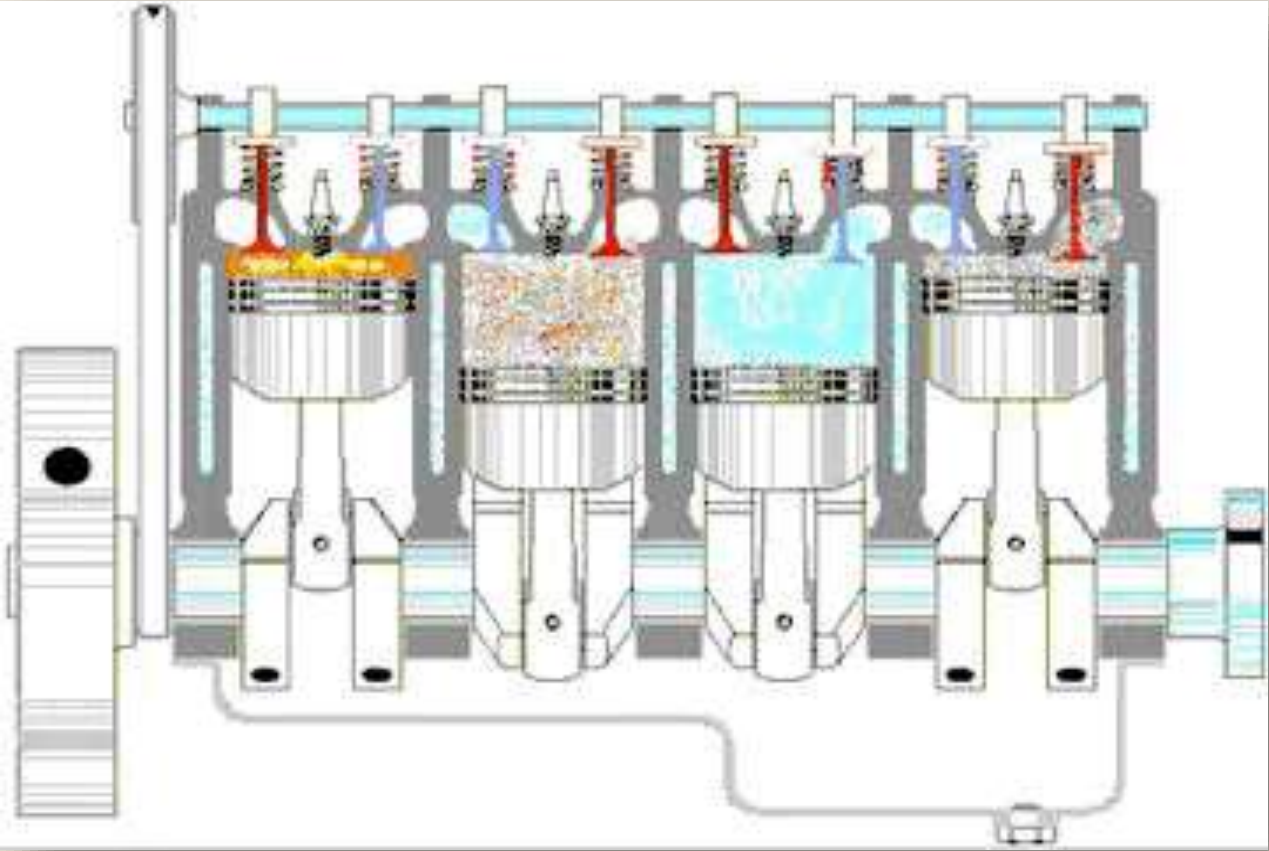


THEORY:

P–V diagram shows variation of cylinder pressure with volume during one engine cycle.

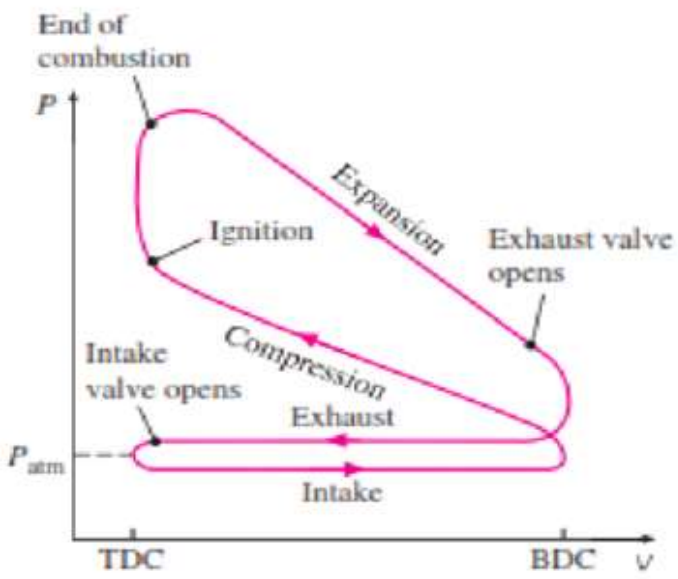
- **Suction (0–1):** Air–fuel mixture enters at nearly atmospheric pressure; volume increases.
- **Compression (1–2):** Charge is compressed adiabatically; pressure rises and volume decreases.
- **Combustion (2–3):** Ignition occurs at nearly constant volume; pressure increases suddenly.
- **Power (3–4):** Adiabatic expansion produces useful work; volume increases and pressure decreases.
- **Exhaust (4–5, 5–0):** Exhaust valve opens and burnt gases are expelled at near atmospheric pressure.



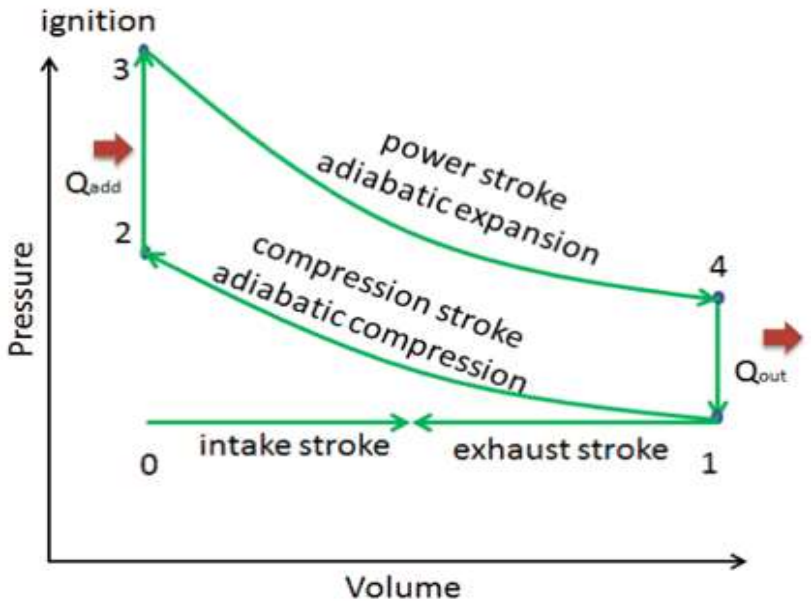


Working of a four-stroke, four cylinder SI Engine (in 1-3-4-2 order)

Ideal vs. Actual PV Diagrams



Actual Otto Cycle



Ideal Otto Cycle

Actual P–V diagram differs from ideal due to pumping losses, heat transfer, and valve timing effects.

APPLICATIONS

1. Determination of work done per cycle
   - The area enclosed by the P–V diagram represents the work done during one engine cycle.
2. Performance analysis of the engine
   - Helps in evaluating engine performance under different operating conditions.
3. Calculation of Indicated Mean Effective Pressure (IMEP)
   - IMEP is obtained from the P–V diagram and is used to assess engine power output.
4. Study of different engine processes
   - Clearly shows intake, compression, combustion, expansion, and exhaust processes.
5. Detection of engine abnormalities
   - Deviations in the diagram indicate problems like poor combustion, leakage, or valve timing issues.
6. Comparison of actual and ideal cycles
   - Used to compare the actual engine cycle with the ideal Otto cycle.
7. Engine design and optimization
   - Helps engineers improve efficiency, compression ratio, and combustion characteristics.

PROCEDURE

1. Check fuel supply, lubrication, and cooling system; ensure all sensor and DAQ connections are proper.
2. Install the pressure transducer on the cylinder head and mount the crank angle encoder; set the TDC reference.
3. Start the engine and allow it to reach steady operating temperature.
4. Set the required engine speed and apply load gradually using the dynamometer.
5. Record cylinder pressure and crank angle data using the DAQ system for multiple cycles.
6. Convert crank angle data into instantaneous cylinder volume using engine geometry.
7. Plot pressure versus volume to obtain the actual P–V diagram of the SI engine.

SIMULATION

CONCLUSION

The P–V diagram of the four-stroke SI engine was obtained successfully, and the different engine processes were identified. The area of the diagram represents the work done by the engine. The actual cycle deviates from the ideal Otto cycle due to heat losses, friction, and valve timing effects, resulting in non-isentropic compression and expansion.

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Thank you

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